

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE:MLA02

COURSE NAME: Fundamentals Of Machine
Learning

COURSE OUTCOME COVERED

CO1: Analyse machine learning fundamentals and mathematical foundations, including probability theory, Bayes decision theory, and information theory. (BL3)

Assessment Tool Weightage:40%

QUESTIONS: 10
Marks:50

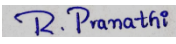
Total

Annexure A

Application-Based Questions

Code of Conduct:

*I R. Pranathi(192525076) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: 

Annexure A

Application-Based Questions

Q1 Implement the concept of probability distribution to predict the likelihood of spam emails in a mail filtering system. (5 Mark- 6 Min)

Q2 Illustrate how Bayes Decision Theory can be applied in medical diagnosis for disease prediction. (5 Mark- 6 Min)

Q3 Develop the design steps of a learning system to build a machine learning solution for student performance prediction. (5 Mark- 6 Min)

Q4 Model the outcome of a weather forecasting problem using random variables and probability theory. (5 Mark- 6 Min)

Q5 Measure the uncertainty in a decision-making dataset by applying information theory concepts. (5 Mark- 6 Min)

Q6 Apply conditional probability concepts to design a fraud detection model for suspicious banking transactions. (5 Mark- 6 Min)

Q7 Demonstrate the use of Bayes theorem in predicting customer purchase behavior from prior shopping history. (5 Mark- 6 Min)

Q8 Construct a machine learning workflow to classify handwritten digits by applying the stages of a learning system. (5 Mark- 6 Min)

Q9 Simulate a real-world stock market prediction problem using random variables and suitable probability distributions. (5 Mark- 6 Min)

Q10 Quantify the uncertainty reduction achieved after feature selection using information gain in a classification dataset. (5 Mark- 6 Min)

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

1ANS:

A probability distribution describes the likelihood of different possible outcomes. In a mail filtering system, probability can be used to determine whether an incoming email is likely to be spam or not spam.

Suppose a mail system has collected 100 previous emails.

Email Category	Number of Emails	Probability
Spam	40	0.40
Not Spam	60	0.60
Total	100	1.00

$$P(\text{Spam}) = \frac{40}{100} = 0.40$$

$$P(\text{Not Spam}) = \frac{60}{100} = 0.60$$

consider that the incoming email contains the word "**Offer**".

Suppose the word "Offer" appears in:

- 30 out of 40 spam emails
- 6 out of 60 non-spam emails

Therefore,

$$P(\text{Offer}|\text{Spam}) = \frac{30}{40} = 0.75$$

and

$$P(\text{Offer}|\text{NotSpam}) = \frac{6}{60} = 0.10$$

The word "Offer" is therefore much more strongly associated with spam emails.

Using Bayes theorem, the probability that an email containing "Offer" is spam can be calculated as:

$$P(\text{Spam}|\text{Offer}) = \frac{P(\text{Offer}|\text{Spam})P(\text{Spam})}{P(\text{Spam})}$$

$$\begin{aligned} P(\text{Offer}) &= (0.75)(0.40) + (0.10)(0.60) \\ &= 0.30 + 0.06 = 0.36 \end{aligned}$$

$$\begin{aligned} P(\text{Spam} | \text{Offer}) &= \frac{0.75 \times 0.40}{0.36} \\ &= \frac{0.30}{0.36} = 0.8333 \end{aligned}$$

So, the probability that the email is spam is approximately 83.33%

Working of the system:

Incoming Email → Extract Words/Features → Calculate Probabilities → Compare Spam and Non-Spam Probability → Spam/Not Spam

Therefore, Probability distribution helps a mail filtering system identify spam by studying the probability of words or features occurring in previous spam and non-spam emails. In this example, an email containing the word "Offer" has an approximately 83.33% probability of being spam.

2ANS:

Bayes Decision Theory provides a mathematical method for making decisions when there is uncertainty. In medical diagnosis, it can be used to predict whether a patient has a disease based on symptoms or medical test results.

Consider a disease called **Disease X**.

Assume:

- Probability of Disease X = 10%
- Probability of a positive test when disease is present = 90%
- Probability of a positive test when disease is absent = 5%

Let:

D = Disease

T = Positive Test

The prior probabilities are:

$$P(D) = 0.10$$

$$P(\neg D) = 0.90$$

Also,

$$P(T|D) = 0.90$$

$$P(T|\neg D) = 0.05$$

Bayes theorem is:

$$P(D|T) = \frac{P(T|D)P(D)}{P(T|D)P(D) + P(T|\neg D)P(\neg D)}$$

Substituting the values:

$$\begin{aligned}P(D|T) &= \frac{(0.90)(0.10)}{(0.90)(0.10)+(0.05)(0.90)} \\&= \frac{0.09}{0.09+0.045} \\&= \frac{0.09}{0.135} \\&= 0.667\end{aligned}$$

Therefore,

$$P(D|T) = 66.7\%$$

This means that after receiving a positive test result, the probability of having the disease is approximately **66.7%** under these assumptions.

Application in medical diagnosis:

1. Collect the patient's symptoms and test results.
2. Calculate the prior probability of the disease.
3. Calculate the probability of the observed test result.
4. Apply Bayes theorem.
5. Compare possible diagnoses.
6. Select the diagnosis with the highest appropriate probability or lowest expected decision cost.

Therefore, Bayes Decision Theory helps medical systems make decisions by combining previous knowledge with new medical evidence. It can support doctors in disease prediction, although actual clinical diagnosis requires appropriate medical testing and professional judgment.

3ANS:

A machine learning system for student performance prediction can predict a student's future academic performance using previous academic and behavioural information.

1. Define the problem

The first step is to clearly identify the objective.

For example:

"Predict whether a student will have Low, Medium or High academic performance."

2. Collect the data

Student information can be collected from academic records.

Possible attributes are:

- Attendance percentage
- Internal examination marks
- Assignment marks
- Previous semester marks
- Study hours
- Number of subjects
- Previous grades

Example:

Attendance	Internal Marks	Study Hours	Previous Mark	Performance
90	85	4	82	High
75	65	2	68	Medium
55	45	1	48	Low

3. Preprocess the data

The collected data may contain missing or incorrect values. Therefore:

- Remove duplicate records.
- Handle missing values.
- Correct incorrect values.
- Convert categorical values into numerical form if required.

4. Select important features

Important features are selected based on their relationship with student performance.

For example, attendance, internal marks and previous marks may be useful features.

5. Split the dataset

The dataset can be divided into:

- Training data
- Testing data

For example:

80% = Training

20% = Testing

6. Select the machine learning algorithm

A suitable algorithm such as:

- Decision Tree
- K-Nearest Neighbour
- Logistic Regression

can be selected.

7. Train the model

The training data is given to the algorithm. The model learns the relationship between student features and performance.

8. Test and evaluate

The trained model is tested using unseen student records.

Performance can be measured using:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \times 100$$

9. Predict a new student's performance

For example, if a new student has:

- Attendance = 92%
- Internal marks = 88
- Study hours = 4
- Previous mark = 85

the trained model may predict:

Performance = High

Workflow:

Data Collection → Data Preprocessing → Feature Selection → Train/Test Split → Algorithm Selection → Training → Testing → Evaluation → Prediction

Therefore, A student performance prediction system can help teachers identify students who are performing well and students who may require additional academic support.

4ANS:

Weather conditions are uncertain because the exact weather for the next day cannot always be known. **Random variables and probability theory** can be used to represent and predict possible weather conditions.

Let XX be a random variable representing tomorrow's weather.

$$X = \{\text{Sunny, Rainy, Cloudy}\}$$

Suppose historical weather data gives the following probabilities:

Weather Condition	Probability
Sunny	0.50
Rainy	0.30
Cloudy	0.20

The total probability is:

$$\begin{aligned} P(\text{Sunny}) + P(\text{Rainy}) + P(\text{Cloudy}) &= 0.50 + 0.30 + 0.20 \\ &= 1 \end{aligned}$$

Therefore, this is a valid probability distribution.

Prediction

The probability of rain tomorrow is:

$$P(\text{Rainy}) = 0.30$$

Therefore, based only on this simplified distribution, there is a:

30%

chance of rain.

Similarly:

$$P(\text{Sunny}) = 50\%$$

$$P(\text{Cloudy}) = 20\%$$

The weather prediction system can use additional random variables such as:

- Temperature
- Humidity
- Wind speed
- Atmospheric pressure
- Previous weather conditions

For example, if humidity increases and atmospheric pressure decreases, the probability of rain may increase.

Expected value

If we assign a numerical value to each weather condition, probability theory can also be used to calculate expected outcomes.

The basic expected value formula is:

$$E(X) = \sum x_i P(x_i)$$

This allows the system to represent the average expected outcome when numerical values are assigned.

Weather forecasting model:

Historical Weather Data → Random Variables → Probability Distribution → Calculate Probabilities → Predict Weather

Therefore, Random variables provide a way to represent possible weather outcomes, while probability theory measures the likelihood of each outcome. Thus, these concepts provide a mathematical foundation for weather forecasting.

5ANS:

Information theory is used to measure the amount of uncertainty or randomness in data. In machine learning, one of the most important concepts from information theory is **entropy**.

Entropy measures the uncertainty in a dataset.

The formula is:

$$H(S) = - \sum_{i=1}^n P_i \log_2(P_i)$$

Suppose a dataset contains **10 students**:

- 6 students passed.
- 4 students failed.

Therefore:

$$P(\text{Pass}) = \frac{6}{10} = 0.6$$

$$P(\text{Fail}) = \frac{4}{10} = 0.4$$

calculate entropy:

$$H(S) = - (0.6 \log_2(0.6) + 0.4 \log_2(0.4))$$

Using:

$$\log_2(0.6) \approx -0.737$$

$$\log_2(0.4) \approx -1.322$$

Therefore:

$$H(S) = - [(0.6)(-0.737) + (0.4)(-1.322)]$$

$$H(S) = 0.442 + 0.529$$

$$H(S) \approx 0.971$$

Therefore, the uncertainty in the dataset is approximately **0.971 bits**.

Interpretation:

- **Entropy = 0** → No uncertainty.
- **Higher entropy** → More uncertainty.
- **Lower entropy** → Less uncertainty.

For example, if all 10 students passed:

$$P(\text{Pass}) = 1$$

and

$$P(\text{Fail}) = 0$$

the entropy would be:

$$H(S) = 0$$

because there is no uncertainty.

Application:

Entropy is especially useful in **decision tree algorithms**. A decision tree selects features that can reduce uncertainty and produce purer groups.

Therefore, Information theory provides a mathematical method to measure uncertainty. In this example, the entropy of the student dataset is approximately **0.971 bits**, showing that the dataset contains a significant amount of uncertainty.

6ANS:

Fraud detection is an important application of machine learning in banking. Conditional probability can be used to calculate the probability that a transaction is fraudulent when certain suspicious characteristics are observed.

Let:

F = Fraudulent Transaction

S = Suspicious Transaction

To find:

$$P(F|S)$$

The conditional probability formula is:

$$P(F|S) = \frac{P(F \cap S)}{P(S)}$$

Using Bayes theorem, it can also be written as:

$$P(F|S) = \frac{P(S|F)P(F)}{P(S)}$$

Suppose the following information is available:

$$P(F) = 0.02$$

This means 2% of transactions are fraudulent.

Suppose:

$$P(S|F) = 0.80$$

This means 80% of fraudulent transactions show suspicious characteristics.

$$P(S) = 0.10$$

$$\begin{aligned} P(F|S) &= \frac{(0.80)(0.02)}{0.10} \\ &= \frac{0.016}{0.10} \\ &= 0.16 \end{aligned}$$

Therefore:

$$P(F|S) = 16\%$$

Features that can indicate suspicious transactions:

1. Very high transaction amount.
2. Transaction from an unusual location.
3. Multiple transactions within a short period.
4. Unusual time of transaction.
5. New or unknown device.
6. Sudden change in spending pattern.

Fraud detection workflow:

Transaction Data → Feature Extraction → Identify Suspicious Features → Calculate Fraud Probability → Fraud Score → Normal/Fraud Alert

A bank can set a threshold. For example, transactions with a sufficiently high fraud probability can be sent for additional verification.

Therefore, Conditional probability allows a fraud detection system to estimate the likelihood of fraud based on observed transaction characteristics. It helps banks identify suspicious transactions and reduce financial losses.

7ANS:

Bayes theorem is useful for predicting customer behaviour because it allows a system to update an existing belief when new information is available.

Consider a customer who is searching for a **laptop online**. We want to predict whether the customer will purchase it.

Let:

$$P(\text{Purchase}) = 0.40$$

This means 40% of similar customers purchase the product.

Suppose:

$$P(\text{Search}|\text{Purchase}) = 0.75$$

This means 75% of customers who purchase the laptop had searched for it online.

Also assume:

$$P(\text{Search}) = 0.50$$

Bayes theorem is:

$$P(\text{Purchase}|\text{Search}) = \frac{P(\text{Search}|\text{Purchase})P(\text{Purchase})}{P(\text{Search})}$$

Substitute the values:

$$\begin{aligned} P(\text{Purchase}|\text{Search}) &= \frac{(0.75)(0.40)}{0.50} \\ &= \frac{0.30}{0.50} \\ &= 0.60 \end{aligned}$$

Therefore:

$$P(\text{Purchase}|\text{Search}) = 60\%$$

So, based on the simplified example, the customer has a **60% probability of purchasing** after searching for the product.

Other information that can be used:

A machine learning system can also consider:

- Previous purchases
- Number of product searches
- Products added to cart
- Customer's purchase frequency
- Product category
- Previous browsing history

Customer prediction workflow:

Shopping History → Extract Customer Features → Calculate Prior Probability → Observe New Behaviour → Apply Bayes Theorem → Purchase Probability → Prediction

For example, a shopping website could recommend the laptop or provide additional product information if the predicted purchase probability is high.

Therefore, Bayes theorem helps update the probability of customer purchase based on previous shopping behaviour and new evidence. It can therefore be used for customer behaviour prediction and personalized recommendations.

8ANS:

Handwritten digit classification is a machine learning problem in which the system learns to identify digits such as **0, 1, 2, ..., 9** from handwritten images.

A complete learning system consists of several stages.

1. Define the problem

The objective is to classify an input handwritten image into one of the ten digit classes:

$$\{0,1,2,3,4,5,6,7,8,9\}$$

2. Collect training data

A dataset containing handwritten digit images is collected. Each image has a corresponding label.

For example:

Image → Label 7

3. Preprocess the data

The images are prepared before training.

Common steps include:

- Converting images to grayscale.
- Resizing images.
- Normalizing pixel values.
- Removing unnecessary noise.

4. Represent the features

Each image is represented using its pixel values.

For example, an image can be represented as:

$$X = [x_1, x_2, x_3, \dots, x_n]$$

where each value represents a pixel.

5. Split the dataset

The data can be divided into training and testing sets.

For example:

80% → Training

20% → Testing

6. Select and train the model

A suitable classification algorithm is selected. The model learns patterns associated with different digits.

Possible algorithms include:

- KNN
- Decision Tree
- Neural Network

7. Test the model

An unseen handwritten digit is provided to the trained model.

For example:

Input: Handwritten image of 7

Model Prediction: 7

8. Evaluate performance

The accuracy can be calculated as:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \times 100$$

Workflow:

Handwritten Digit Dataset



Preprocessing



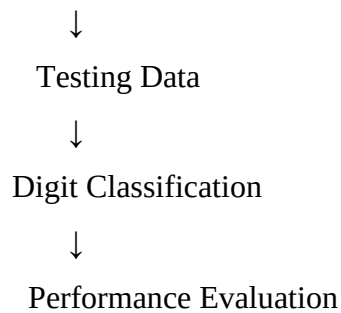
Feature Extraction



Training Data



Machine Learning Model



Therefore, The learning system learns patterns from labelled handwritten images and uses these patterns to classify new digits. Such a system can be used in applications such as **automatic form reading, postal code recognition and digit recognition.**

9ANS:

Stock market prices are affected by many uncertain factors. Therefore, probability theory and random variables can be used to represent possible changes in stock prices.

Let:

X = Daily percentage change in stock price

Suppose historical data is used to classify the daily movement into three categories:

Stock Movement	Probability
Decrease	0.30
No major change	0.20
Increase	0.50

The total probability is:

$$0.30 + 0.20 + 0.50 = 1$$

Therefore, it forms a valid probability distribution.

Probability of increase:

$$P(\text{Increase}) = 0.50$$

Therefore, based on this simplified historical model, there is a 50% probability of an increase.

Probability of decrease:

$$P(\text{Decrease}) = 0.30$$

Therefore, there is a 30% probability of a decrease.

Random variables used in stock prediction

A stock prediction model can use several variables:

- Previous stock price
- Daily return

- Trading volume
- Market index
- Interest rate
- Company performance
- Historical price movement

For example, the daily return can be calculated as:

$$\text{Return} = \frac{\text{Today's Price} - \text{Yesterday's Price}}{\text{Yesterday's Price}}$$

The historical returns can then be represented using a suitable probability distribution. A normal distribution is often used as a simple approximation for modelling continuous returns, although real financial data may not follow it perfectly.

Stock prediction workflow:

Historical Stock Data → Calculate Returns → Define Random Variables → Estimate Probability Distribution → Analyse Current Data → Predict Possible Movement

Example:

Suppose the current stock is ₹500 and the historical model predicts an increase with probability 0.50. The system can classify the next movement as:

Most likely category: Increase

This does not guarantee the actual market outcome because stock prices are influenced by many unpredictable factors.

Conclusion:

Random variables and probability distributions provide a mathematical framework for modelling uncertainty in stock markets. They can help estimate the probability of possible price movements and support decision-making.

10ANS:

Information Gain is an important concept in information theory used for feature selection. It measures how much uncertainty is reduced when a particular feature is used to divide a dataset.

$$\text{IG}(S,A) = H(S) - H(S|A)$$

where:

- $\text{IG}(S,A)$ = Information Gain of feature A
- $H(S)$ = Entropy of the original dataset

- $H(S|A)$ = Entropy after using feature A

Suppose a classification dataset contains students who are either **Pass or Fail**.

Initially:

- 6 students = Pass
- 4 students = Fail

Therefore:

$$H(S) = 0.971$$

Now suppose we select the feature **Attendance**.

After dividing the dataset based on attendance, assume the weighted entropy is:

$$H(S|Attendance) = 0.571$$

Therefore:

$$\begin{aligned} IG(S, Attendance) &= H(S) - H(S|Attendance) \\ &= 0.971 - 0.571 \\ IG &= 0.400 \end{aligned}$$

Thus, the uncertainty is reduced by **0.400 bits** after using the Attendance feature.

Interpretation:

A feature with higher information gain provides more useful information for classification.

For example:

Feature	Information Gain
Attendance	0.400
Study Hours	0.250
Age	0.080

Here, **Attendance** has the highest information gain, so it would be preferred over the other features in this simplified example.

Relation to Decision Tree:

Decision tree algorithms such as ID3 calculate information gain for available features and generally choose the feature with the highest information gain for splitting the data.

Workflow:

Original Dataset → Calculate Entropy → Select Feature → Calculate Conditional Entropy → Calculate Information Gain → Select Best Feature

Therefore, Information gain quantitatively shows how much uncertainty is removed by selecting a feature. In this example, Attendance reduces uncertainty by 0.400 bits, making it the most useful feature among the compared features